

EGCSE Study Room Project Plan

MVP requirements and development proposal

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Project stage Concept and planning

Project overview

We propose to develop EGCSE Study Room, a browser-based learning platform for Mathematics and Physical Science learners in Eswatini. It will bring syllabus-based lessons, worked examples, past-paper references and related learning resources into a clear chapter-by-chapter study process.

The first release will be a minimum viable product, or MVP: a usable version that allows learners and teachers to complete the main study activities and provide feedback. Learners will study a topic, attempt a question, revisit a prerequisite when needed and practise a similar question. They will also be able to prepare a question for an external AI chat and save a checked explanation as a PDF.

The proposed delivery approach will keep the website static so that it can be hosted through a public GitHub repository without a paid application server. Progress and uploaded papers will initially be stored in the learner's browser. Downloadable copies and explicit exports will support study where internet access is limited.

Expected value

- Give learners a clear route from a syllabus topic to an explanation and a practice task.
- Reduce time spent searching for the prerequisite concept needed at a particular step.
- Help teachers prepare focused practice and share suitable papers or study materials.
- Keep the first release affordable to operate and practical to maintain.

These are intended benefits to investigate through a pilot. The plan does not assume that learning gains, user demand or technical performance have already been demonstrated.

Plan contents

Planning area	Pages
Problem definition, users and objectives	2
MVP scope and functional requirements	3
Proposed learner and teacher journeys	4
Curriculum and content development	5
Technical approach and quality requirements	6
Delivery schedule, responsibilities and resources	7
Testing, pilot targets and release criteria	8
Risks, decisions and planning references	9

Problem users and objectives

Problem statement

A learner working through an exam question may understand the topic but become stuck on a smaller prerequisite, such as fractions, rearranging a formula or Pythagoras. Searching for help can interrupt the task, while a complete answer alone may not explain the missing concept. Lessons, videos and past papers may also be spread across different websites.

The project will address this proposed problem by putting a short lesson, a worked example, targeted refresher links and practice questions in one study journey. Initial interviews will check whether these difficulties are common in the intended learner group and which ones should receive priority.

Intended users

User group	Need the MVP should meet
EGCSE learners	Find the right topic, understand a method, practise independently and keep useful work.
Teachers and tutors	Guide revision, identify suitable paper questions and share study materials with learners.
Project owner and content reviewers	Maintain the syllabus mapping, check explanations and correct problems efficiently.

These will be usage roles. Separate student and teacher accounts will not be required for the first release.

Project objectives

- 1. Organise the course.** Map the agreed Mathematics and Physical Science syllabus topics into a searchable chapter structure, with the relevant Mathematics pathway clearly identified.
- 2. Support understanding.** Give each published chapter a learning objective, explanation, worked example, prerequisite links and an end-of-chapter task.
- 3. Connect study to exams.** Show the subject, paper, session, question and page for every verified past-paper reference. Label original practice separately.
- 4. Enable further practice.** Generate similar questions with consistent difficulty, valid values and matching worked answers after a learner engages with the first task.
- 5. Retain useful learning.** Support PDF study sheets, local progress records and explicit backups or class exports.

Discovery before development

The project lead will seek feedback from approximately five learners and two teachers before fixing the design. Discussion will cover devices, connectivity, difficult topics, preferred explanations and how papers are currently shared. These numbers are recruitment targets, not confirmed participants.

The initial assumptions are that learners can access a modern browser, teachers can help review subject content, and free static hosting will be sufficient. Discovery will test these assumptions and define the exam cohort and syllabus edition.

MVP scope and functional requirements

The MVP release target will cover the agreed topic headings in both subjects with selected revision skills. A smaller internal prototype will be used first to test the learning process. Published coverage will be stated clearly; one example per topic will not be presented as a complete textbook.

ID	Required capability	Priority
F01	Browse and search subjects and chapters; identify the applicable Mathematics pathway.	Must
F02	Present a chapter objective, explanation, worked example, practice task and model explanation.	Must
F03	Place a relevant book, article or video link beside a step that uses a prerequisite. Explain the link on hover or keyboard focus.	Must
F04	Link end-of-chapter exam tasks to a verified paper, year, session, question and PDF page where available.	Must
F05	Unlock similar practice after an attempt or opened working; generate a different valid question with a matching solution.	Must
F06	Add PDF papers to a local collection, label them and export selected papers in a downloadable class copy.	Must
F07	Prepare an AI study prompt, link to an external chat and accept the pasted explanation for formatting.	Must
F08	Save a chapter, practice question or formatted solution through the browser's PDF print option.	Must
F09	Keep chapter notes and progress locally, with export, import and reset controls.	Must
F10	Provide a downloadable version containing the core lessons, question templates and required app assets.	Must
F11	Let a teacher add a chapter reference and share those references separately from personal notes.	Should

Must requirements will be release conditions. A Should requirement may move to the next release only through a recorded scope decision. Each capability will be tested against the acceptance criteria on page 8.

Features reserved for later phases

Automatic AI responses, central user accounts, shared cloud storage, teacher dashboards, class analytics, automated reading of scanned papers, payments and additional subjects will be outside the initial scope. Adding these would require further design, funding or services beyond static hosting.

MVP feasibility boundary

Uploading a paper will initially add it to that browser, not to a shared website database. Class sharing will use an explicit export. AI assistance will initially use a guided copy-and-paste workflow; automatic transfer of questions and returned answers will be considered after the MVP has been evaluated.

Proposed user journeys

Learner studying a chapter

- 1. Choose the topic.** The learner will select a subject, choose the relevant pathway where needed, and open a chapter.
- 2. Learn the idea.** The chapter will introduce its objective, key explanation, useful formula and common mistake.
- 3. Attempt the example.** The learner will try the example before opening its step-by-step working. A step using a prerequisite will name the concept and provide a focused refresher link.
- 4. Attempt an exam skill.** The learner will open a referenced paper question or complete the clearly labelled original practice task at the end of the chapter.
- 5. Check and repeat.** After an attempt or review of worked steps, the learner will be able to generate a similar question. Its values, answer and explanation will update together.
- 6. Record the next step.** The learner will write a useful note, mark the chapter reviewed and retain a backup when needed.

Example of a targeted refresher

A trigonometry example might require the learner to rearrange a formula before calculating an unknown side. The words Rearranging a formula will be linked beside that step. Hovering or focusing the link will show Click to learn rearranging a formula. Selecting it will open a relevant resource while keeping the chapter available. On a touch screen, the link will remain directly usable.

Teacher preparing a revision activity

The teacher will select a chapter and check its reference question. They will be able to add suitable PDFs to a local collection and export selected papers for learners. A separate reference export is proposed so that paper links can be shared without including personal notes or answers.

Learner requesting AI help

The learner will choose a question, include their own attempt and select either hints or full working. The app will prepare a prompt to copy into ChatGPT or Gemini. Where necessary, the learner will attach the scan in that external chat. They will paste the response back, check it and save a formatted PDF study sheet. No automatic return of the response is planned for the MVP.

Interface principles

- Use plain labels, readable equations and a clear sequence within every chapter.
- Reveal worked answers deliberately so that learners have an opportunity to try first.
- Show only the refresher relevant to the current step, rather than a long unrelated resource list.
- Provide useful feedback for invalid answers, missing files and unavailable browser storage.

Curriculum and content development

Confirm the syllabus before writing

The curriculum audit will use the supplied Khanyisa syllabus collection as a starting point and confirm the applicable ECESWA requirements for the intended exam cohort. Mathematics 6880 and Physical Science 6888 are the proposed subject scope. The 2024-2026 material will inform the initial audit, but a later cohort must use its applicable edition. [1]

The audit will define the chapter list, Mathematics pathways, learning objectives and any practical skills that require teacher-led activity. Content will then be written in batches. A small prototype covering Mathematics, Chemistry and Physics will be reviewed before the same structure is applied to the remaining topics.

Planned chapter structure

Element	Content to prepare
Learning objective	A specific skill and its syllabus reference, with any pathway restriction.
Short lesson	A clear explanation, key terms, relevant formula and common misconception.
Worked example	Original step-by-step working with assumptions, units and checked calculations.
Refresher links	A small set of relevant links attached to the steps that need them.
End-of-chapter tasks	An original practice task, an explanation task and a verified paper reference where available.
Similar practice	A question template with controlled values, a valid answer and consistent worked steps.

Source and review process

- 1. Record sources.** Maintain a register containing topic, syllabus edition, provider, URL, paper identifiers, question/page, review date and reviewer.
- 2. Write original teaching material.** Prepare explanations and examples at the appropriate EGCSE level. Link to publisher papers instead of assuming they can be republished in full.
- 3. Check independently.** A subject reviewer will check the method, answer, units, difficulty and mapping. Unresolved errors will keep the affected material out of release.
- 4. Label gaps clearly.** If no suitable paper question has been verified, label the task as original practice and record the missing reference. A paper number will never be invented.

Select supplementary materials

Books, articles, videos and simulations will be selected for accuracy, clarity, syllabus relevance and ease of access. JS Maths Institute is a suggested provider; the exact channel and individual videos will be confirmed before use. Links will be reviewed periodically, with a replacement recorded when a resource moves or becomes unsuitable.

Technical approach and quality requirements

We propose HTML, CSS and JavaScript for a static website served through GitHub Pages. This service supports static project sites from GitHub repositories and is available for public repositories on GitHub Free. [2] The design will avoid a paid server dependency for the core learning features.

Component	Proposed responsibility
Browser interface	Subject navigation, chapters, accessible controls, responsive layout and print views.
Content catalogue	Structured lessons, syllabus references, paper metadata and supplementary links.
Question engine	Controlled templates, variant identifiers, answer checking and matching worked solutions.
Local storage	Small progress and draft records in localStorage; uploaded PDF files in IndexedDB.
Export tools	Progress backups, offline app copies, class paper packages and browser-based PDF output.
External services	Publisher resources and learner-selected AI chats, opened through explicit links.

Proposed data model

The main records will be Subject, Chapter, LearningResource, PaperReference, QuestionTemplate, PracticeSession and UploadedPaper. Each chapter will have a stable identifier. Paper references will store subject code, year, session, paper, question, page and URL. Backups will include a format version so future changes can be handled safely.

Quality requirements

- **Accessibility:** core study tasks will support keyboard use, visible focus, readable labels and touch interaction. Tooltips will supplement usable links.
- **Mobile use:** layouts will be checked from 320 pixels wide through desktop sizes, including equation wrapping and large touch targets.
- **Performance:** target a usable first chapter within five seconds on an agreed pilot device and connection; record the test conditions.
- **Offline use:** core lessons and generated practice will work in the downloaded copy. External resources and AI services will require internet access.
- **Reliability:** show storage errors, validate backup imports and provide exports before learners change devices or clear data.
- **Data handling:** class exports will exclude personal notes and AI drafts. Pasted responses will be treated as text, with controlled equation formatting rather than executable HTML.

Proposed starting limits are 20 MB per PDF and 60 MB per paper collection, subject to device testing. Browser data will not automatically synchronise across devices. No paid-service API keys will be placed in public app files; a later automatic AI service would require a secure backend and a separate cost decision.

Delivery schedule and resources

Proposed development method

Work will proceed in short iterations: design a study task, build it, check it with reviewers and adjust it before expanding. The following 12-week schedule is a planning estimate from an agreed start date. It assumes one developer contributing roughly 15-20 hours per week and regular access to Mathematics and Physical Science reviewers.

Period	Work and intended output	Stage condition
Weeks 1-2	Confirm learner needs, syllabus edition, chapter map, content rights and requirements.	Scope and review responsibilities agreed.
Weeks 3-4	Create wireframes and a small prototype covering a topic in each subject area.	Learners can follow the proposed study journey.
Weeks 5-8	Build the MVP features and prepare chapter content in parallel batches.	Must requirements implemented for review.
Weeks 9-10	Finish the agreed topic coverage, check references and review generated answers.	Content checklist complete; major issues resolved.
Week 11	Run the supervised pilot, mobile/offline checks and export tests.	Acceptance evidence and feedback recorded.
Week 12	Correct release issues, prepare documentation and publish the approved version.	Release checklist accepted by the project lead.

Content review will run throughout development. Reviewer availability, scope and pilot findings may change the dates. Any reduced release scope will require a recorded project decision.

Responsibilities

Mpumelelo Mancoba Dlamini is proposed as project lead and developer. Mathematics and Physical Science reviewers will be recruited to approve content, while teachers will help arrange learner feedback. The project lead will maintain the issue list and source register. Supporting appointments are still to be confirmed.

Budget assumptions

Resource	Planning basis
Static hosting	Target E0 recurring hosting with an eligible public GitHub Pages repository and default address. [2]
Development and review	Estimate 180-240 developer hours. Agree reviewer time and fees separately.
Pilot costs	Use available devices. Confirm data, travel and printing costs before recruitment.
AI services	No paid AI API in the MVP. External chat access will follow each provider's terms.

A zero hosting target does not mean zero project cost. Labour, connectivity and any agreed reviewer fees must be budgeted before the schedule is committed.

Validation and acceptance criteria

The checks below are planned release conditions. No test results or learner outcomes are assumed. The project lead will keep a requirements checklist linking each feature to its evidence, reviewer and unresolved issues.

Area	Proposed acceptance evidence
Course navigation and content F01-F02	Every agreed chapter is reachable, correctly labelled and reviewed. Each has the required lesson, example and tasks.
Refresher resources F03	Each published link is relevant and opens correctly; its explanation is available by keyboard as well as hover.
Paper references F04	Every claimed original-paper reference has been checked against its PDF. Missing references are clearly identified.
Question variants F05	Check at least 100 variants per template, including boundary cases. Independently verify representative answers and ensure the next variant differs.
Paper sharing F06	A selected PDF opens after reload and in an exported class copy on another test browser. Oversized or invalid uploads show a useful message.
AI study sheets and printing F07-F08	The prompt preserves the exact question and attempt. A pasted response formats safely, and saved PDFs show readable equations and complete text.
Progress and offline use F09-F10	A backup restores the intended data. Imports warn before replacement. Downloaded lessons and practice work with the network disabled.
Teacher references F11	If included, exported links import correctly without carrying personal notes or answers.
Cross-cutting quality	Core tasks work on the agreed phone and desktop browsers, with no blocking errors, clipped controls or unintended data in shared exports.

Pilot design and targets

The proposed pilot will involve about ten learners and two teachers, subject to recruitment. Learners will be asked to find a chapter, use a refresher, attempt a question, generate similar practice and save a study sheet. Teachers will check whether the materials fit their revision needs.

- Usability target: at least eight of ten learners complete the main chapter-to-practice task without the facilitator taking over.
- Comprehension check: learners explain a worked method and try a related question; record misconceptions to improve content rather than claiming exam-grade gains.
- Release gate: all Must requirements accepted, all published content reviewed, and no unresolved issue that blocks core study or produces a known incorrect answer.

A short pilot can guide improvements but will not establish long-term learning impact. Findings will determine whether to release, revise the content or run another focused test.

Risks decisions and next steps

Risk or dependency	Planned response
Wrong syllabus edition or incomplete coverage	Confirm the exam cohort before content writing; track every agreed topic and review changes before a new cycle.
Incorrect questions or misleading explanations	Use independent subject review, controlled generators and a clear route for reporting a question and variant.
Unavailable or unsuitable external resources	Keep a source register, check links before release and provide replacements when practical.
Paper redistribution restrictions	Link to original publishers by default and confirm permission before packaging papers for distribution.
Loss of locally stored work	Explain device-specific storage, provide explicit exports and handle unavailable storage without hiding the limitation.
Overreliance on AI answers	Prefer hints and learner attempts, keep the response source visible and require checking before a solution is treated as reliable.
Content workload or reviewer delays	Prepare material in reviewable batches, recruit reviewers early and revise the schedule rather than release unchecked content.

Decisions needed before implementation

- 1. Agree the learning scope.** Confirm the target exam cohort, syllabus edition, Mathematics pathways and the chapter coverage required for MVP release.
- 2. Confirm people and resources.** Recruit subject reviewers and pilot teachers, agree their availability and establish any costs.
- 3. Approve the product boundaries.** Confirm local storage, explicit class exports and external chat handoff as the first-release approach.
- 4. Set the first milestone.** Agree a start date, select the prototype topics and prepare wireframes and the requirements checklist.

Planned deliverables

The project will produce the agreed requirements and chapter map, reviewed lessons and question templates, a responsive static app, offline and sharing tools, test and pilot records, student and teacher guidance, and a maintenance guide. Publication will follow acceptance of the release checklist.

Planning references

[1 Khanyisa syllabus collection](#)

[Khanyisa Form 5 past-paper collection](#)

[2 GitHub Pages service and hosting eligibility](#)

The next stage will be requirements confirmation and prototype design. Estimates and technical choices will be reviewed at that stage before development work is committed.